5 Analysis Models

Seven conceptual classes hold everything the analyzed use cases mention. UC-01 and UC-02 are traced through them message by message, each controlled by one interaction object (ApplicationManager, StatusUpdater). Technology stays switched off: no screens, storage or framework names appear below.

5.1 Domain model

InternshipApplication is the center of the model: one StudentAccount keeps many of them, and each one gathers its own interviews, status history, documents and reminders.

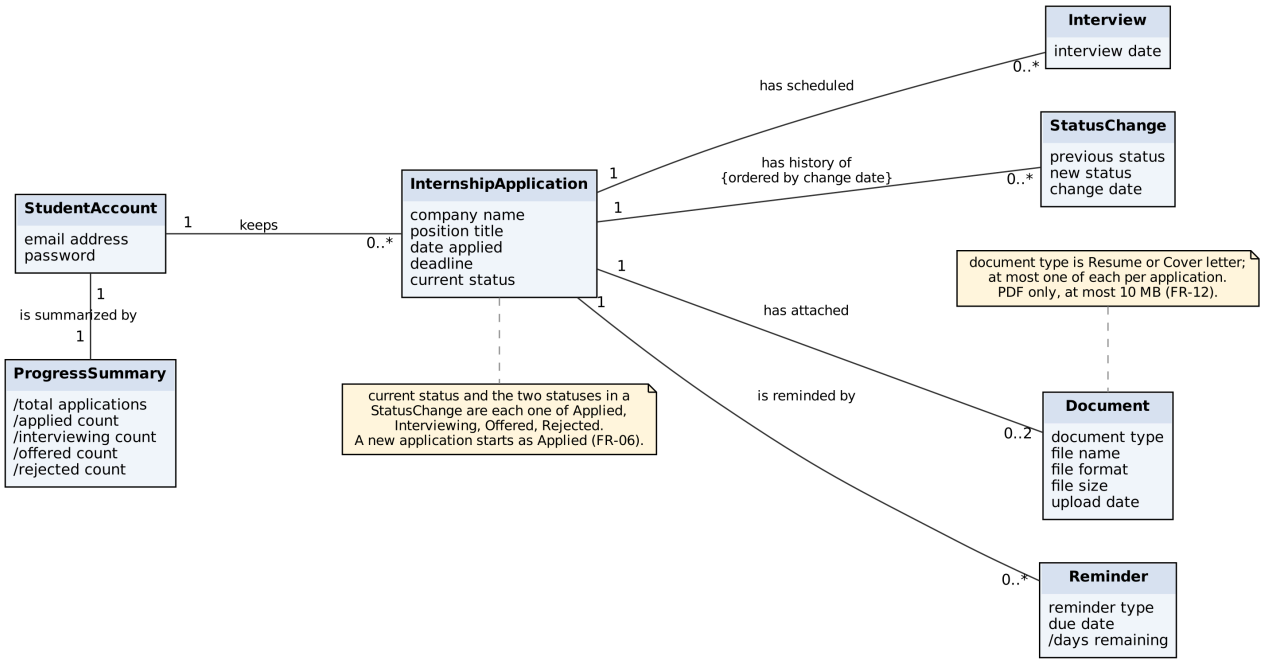


Figure 5.1 Domain model (conceptual classes, attributes, associations with multiplicities)

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
StudentAccount	A student's registered InternTrack account, the owner of every record below it	email address, password	keeps 0..* InternshipApplication; is summarized by 1 ProgressSummary
InternshipApplication	One internship the student has applied to	company name, position title, date applied, deadline, current status	kept by 1 StudentAccount; has scheduled 0..* Interview; has history of 0..* StatusChange; has attached 0..2 Document; is reminded by 0..* Reminder

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
Interview	One interview scheduled for an application	interview date	scheduled for 1 InternshipApplication
StatusChange	One entry in an application's status history	previous status, new status, change date	belongs to 1 InternshipApplication, ordered by change date
Document	A resume or cover letter the student attached to an application	document type, file name, file format, file size, upload date	attached to 1 InternshipApplication
Reminder	An in-app notice for a deadline or interview date 0 to 7 days away	reminder type, due date, /days remaining	about 1 InternshipApplication
ProgressSummary	The student's dashboard counts, worked out from their applications	/total applications, /applied count, /interviewing count, /offered count, /rejected count	summarizes 1 StudentAccount

Rules the model carries (two appear as notes on Figure 5.1):

- current status, previous status and new status each take exactly one of Applied, Interviewing, Offered or Rejected. A new application starts as Applied (FR-06), and a StatusChange never has the same previous and new status (UC-02 A1).
- company name, position title and date applied are required. deadline is optional (FR-03, FR-08).
- document type is Resume or Cover letter, with at most one of each per application. Each is a PDF of at most 10 MB (FR-12).
- Attributes marked / are derived from the student's applications and dates, never entered by anyone (FR-09, FR-10).
- A Reminder is shown only inside InternTrack and never sent out (CON-03).

5.2 Object interaction models

UC-01 and UC-02 are carried through. Together they realize seven of the twelve functional requirements (FR-03 to FR-08 and FR-12). UC-02 holds the riskiest step: keeping an application's status and its history in step. Each diagram adds one interaction object to the classes of 5.1:

Interaction object	What it does (one line)	Used in
ApplicationManager	Controls UC-01: takes the student's entries, runs the checks, and creates, links or removes the record	5.2.1
StatusUpdater	Controls UC-02: fetches the chosen record from the student's own, applies the new status with its history entry, and shows the result	5.2.2

5.2.1 UC-01 Manage Internship Application Record

(a) Non-trivial steps. This analysis follows UC-01 in 4.2: the add flow (0) to (6) and alternate flow A2 (delete).

- (0) needs the student's own records, so ApplicationManager asks StudentAccount for them. Only that student's records appear (QR-02).
- (4) needs the file check (FR-12) and the required-field check (FR-03).
- (5) needs a new InternshipApplication with status Applied, its Interview and Document entries, and the link to StudentAccount (FR-03, FR-06, FR-08, FR-12), then the updated list.
- A2 needs the student's confirmation before StudentAccount gives up the record (FR-05).
- (1) and (2) are trivial: the student picks Add Application and sees an empty entry with no stored data. (3) is the student's input and appears below only as the message that starts (4). (6) is the student reading the list.
- A1 (edit) reuses rows 3 to 5.5: the record's details are shown, the same checks run at (4), and at (5) the existing InternshipApplication is updated instead of created. It is not drawn again.
- 4.1 lists duplicate detection as non-trivial, but 4.2 has no step for it and no requirement asks for it, so it is not modeled here (see 5.3, K5).

(b) Scenario table. Rows starting with A2 belong to alternate flow A2 in 4.2, where the student deletes an application.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
0.1	ApplicationManager	asks for the logged-in student's application records	—	StudentAccount
0.2	ApplicationManager	shows the student's own records and the option to add one	company name, position title and current status of each record	Student (actor)
3	Student (actor)	submits the entered details and files	company name, position title, date applied, deadline, interview dates, resume file, cover letter file	ApplicationManager
4.1	ApplicationManager	checks each file is a PDF of at most 10 MB	file format, file size	ApplicationManager (itself)
4.2	ApplicationManager	checks company name, position title and date applied are filled in	entered details	ApplicationManager (itself)
5.1	ApplicationManager	creates the application record with status Applied	company name, position title, date applied, deadline	InternshipApplication

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
5.2	InternshipApplication	creates an interview entry for each entered interview date	interview date	Interview
5.3	InternshipApplication	attaches each file as a resume or cover letter	document type, file name, file format, file size, upload date	Document
5.4	ApplicationManager	adds the new application to the student's records	the new InternshipApplication	StudentAccount
5.5	ApplicationManager	shows the updated list with the new record marked Applied	updated list of records	Student (actor)
A2.1	Student (actor)	chooses to delete one of the listed applications	the chosen application	ApplicationManager
A2.2	ApplicationManager	asks the student to confirm the deletion	company name and position title of the chosen application	Student (actor)
A2.3	Student (actor)	confirms the deletion (if the student cancels, nothing is removed)	—	ApplicationManager
A2.4	ApplicationManager	asks it to remove the application from the student's records	the chosen InternshipApplication	StudentAccount
A2.5	ApplicationManager	shows the updated list without the deleted record	updated list of records	Student (actor)

(c) Analysis sequence diagram. ApplicationManager is in control. The 15 messages are the 15 rows above, with the same numbers and wording.

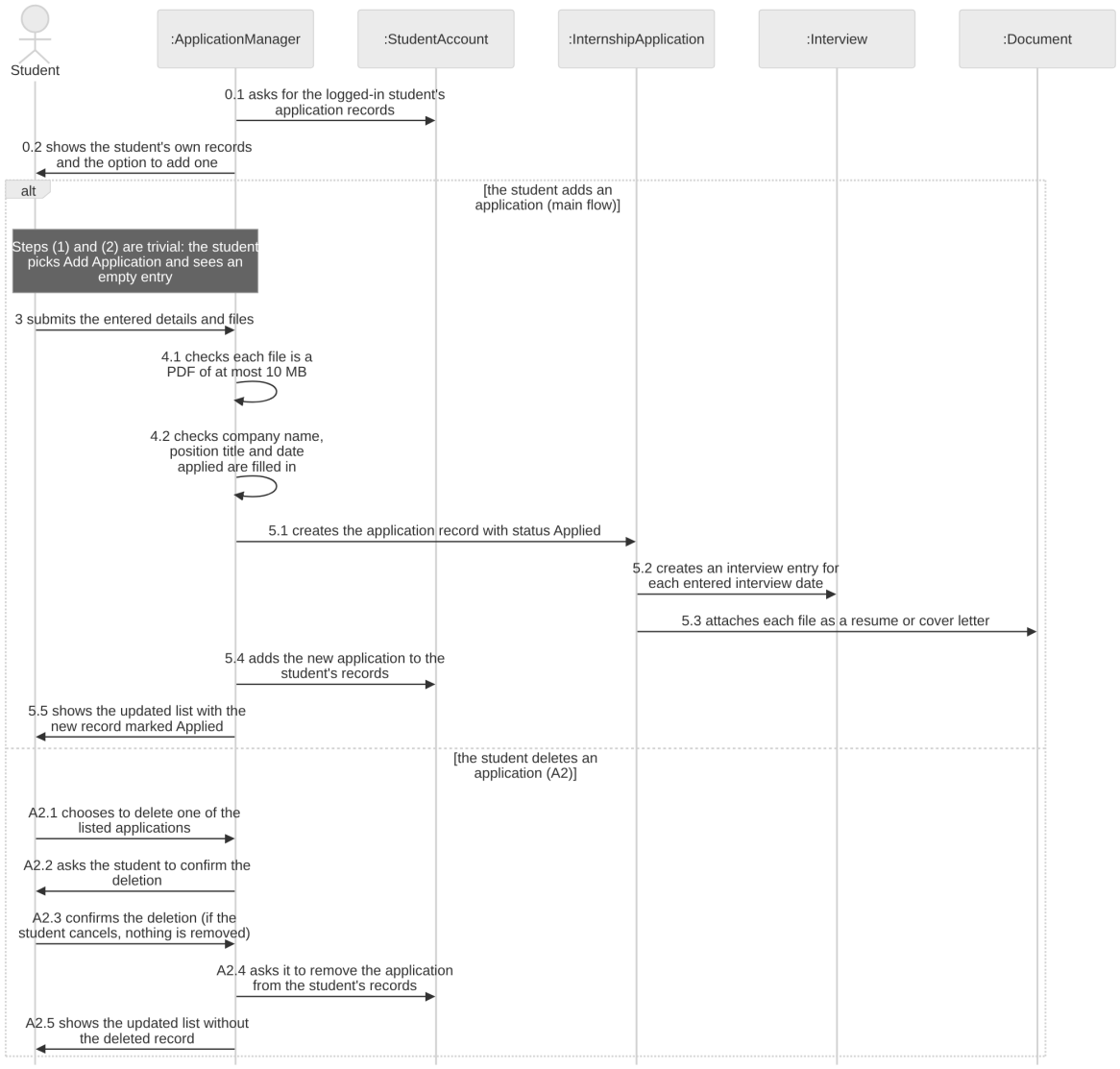


Figure 5.2.1 Analysis sequence diagram for UC-01; message numbers match the scenario table

5.2.2 UC-02 Update Application Status and Track History

(a) Non-trivial steps. This analysis follows UC-02 in 4.2: turns (0) to (7), with A1 (same status chosen, at step 4) and A2 (Interviewing chosen, at step 3).

- (0) is the same interaction as UC-01 rows 0.1 and 0.2 and is not repeated.
- (2) needs the chosen application from the student's own records (QR-02) and its current status and history (FR-07).
- (4) needs the same-status check (FR-06).
- (5) needs a new StatusChange and the new current status, kept together so they never disagree (FR-06, FR-07). In A2 it also needs an Interview entry for each entered date (FR-08).
- (1) and (3) are the student's input and appear below only as the messages that start (2) and (4). (6) shows what (5) produced, and (7) is the student reading it.
- New details other than status at (3) follow the UC-01 edit flow (A1) and are not repeated. Reminders for the new interview dates belong to UC-03.

(b) Scenario table. Rows starting with A1 or A2 belong to those alternate flows in 4.2.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	Student (actor)	selects an application from the list	the chosen application	StatusUpdater
2.1	StatusUpdater	asks for the chosen application from the student's own records	the chosen application	StudentAccount
2.2	StatusUpdater	asks for its current status and status history	—	InternshipApplication
2.3	StatusUpdater	shows the application, its current status and its history in date order	company name, position title, current status, StatusChange entries	Student (actor)
3	Student (actor)	chooses a new status and saves	new status	StatusUpdater
A2.1	StatusUpdater	asks for the interview dates	—	Student (actor)
A2.2	Student (actor)	enters the interview dates	interview dates	StatusUpdater
4	StatusUpdater	checks the new status differs from the current status	current status, new status	StatusUpdater (itself)
A1.1	StatusUpdater	says nothing has changed, no history entry is added	current status	Student (actor)
5.1	StatusUpdater	asks it to change to the new status	new status, change date, interview dates	InternshipApplication
5.2	InternshipApplication	records the previous status, new status and change date	previous status, new status, change date	StatusChange
5.3	InternshipApplication	sets its current status to the new status	new status	InternshipApplication (itself)
A2.3	InternshipApplication	creates an interview entry for each entered date	interview date	Interview

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
6	StatusUpdater	shows the new status and the updated history	new status, StatusChange entries	Student (actor)

(c) Analysis sequence diagram. StatusUpdater is in control. Messages 5.2 and 5.3 happen together or not at all, so the status and its history never disagree. The 14 messages are the 14 rows above, with the same numbers and wording.

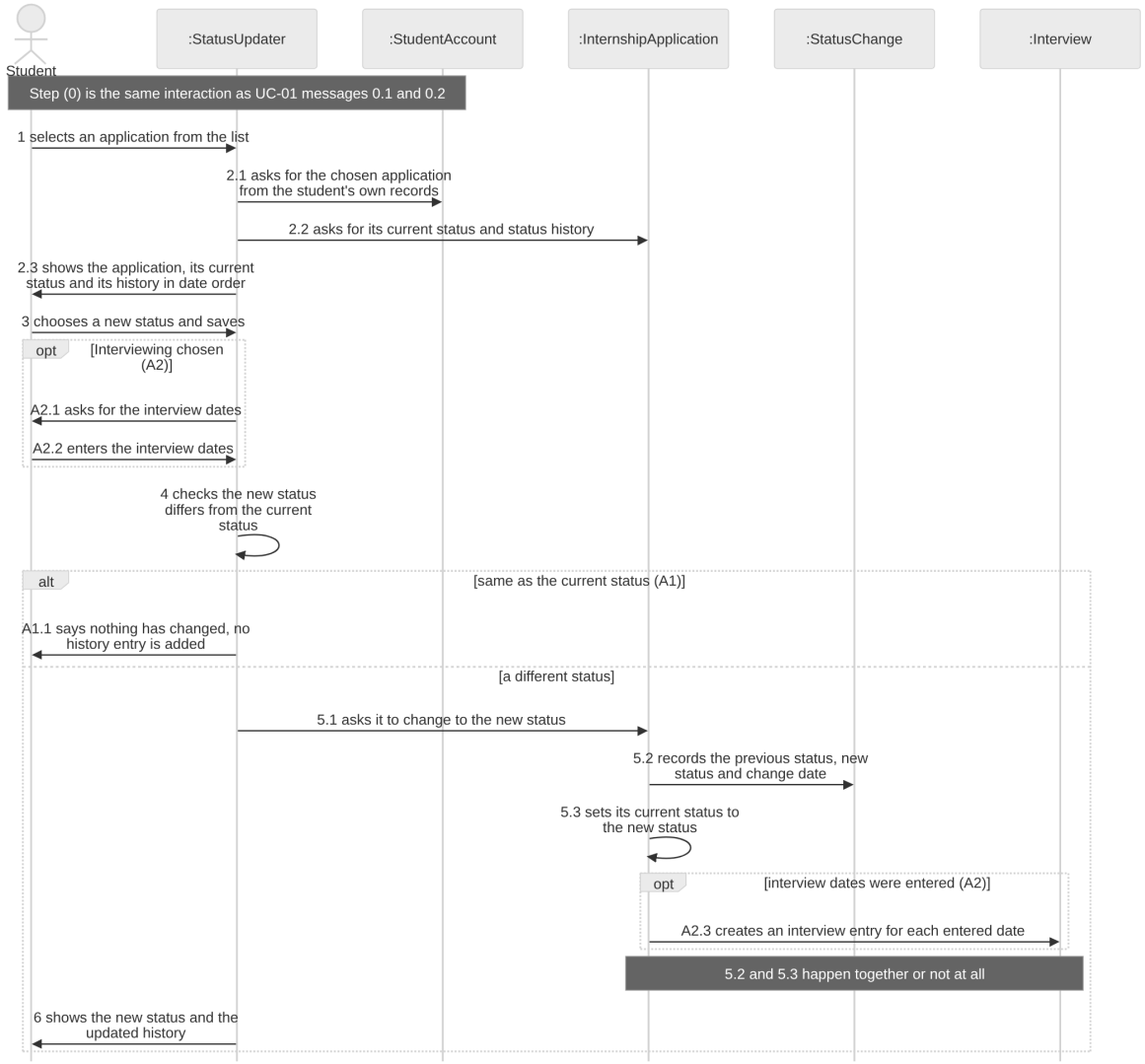


Figure 5.2.2 Analysis sequence diagram for UC-02; message numbers match the scenario table

5.3 Consistency cross-check

Three checks failed on the first pass and were fixed (K1, K2, K4). The other three pass, and K5 found a step with no requirement behind it.

Check	Result (pass, or what failed)	Fixed in
K1 Every item in an actor column (4.2) is an attribute in the domain model (5.1)	Failed, then passed. An earlier draft of UC-01 step (2) asked only for a resume, while FR-12 also allows a cover letter. Step (2) now asks for both, and Document.document type covers both. All other items match: company name, position title, date applied, deadline (InternshipApplication); new status (InternshipApplication, StatusChange); interview dates (Interview); email address and password (StudentAccount, UC-05).	4.2 UC-01 step (2); 5.1 Document
K2 Every object on a sequence diagram is a domain class or a declared interaction object	Failed, then passed. The first draft named the domain class Student, the same as the actor, so the diagrams could not tell the person from the account record. It is now StudentAccount. ApplicationManager and StatusUpdater are declared in 5.2.	5.1, 5.2.1, 5.2.2
K3 Every message on a sequence diagram corresponds to exactly one scenario-table row	Pass. The UC-01 diagram has 15 messages and its table 15 rows; UC-02 has 14 and 14. Numbers and wording match, alternate-flow rows included. An earlier draft worded three UC-01 labels differently in the table and the diagram; they now match.	5.2.1 (b)
K4 Every non-trivial step in 5.2 (a) has at least one scenario-table row	Failed, then passed. After 4.2 was revised (submitting moved into step (3), both checks into (4), saving into (5)), 5.2 still used the old step numbers. Both tables and diagrams were renumbered. Now UC-01: (0) rows 0.1–0.2, (4) rows 4.1–4.2, (5) rows 5.1–5.5, A2 rows A2.1–A2.5. UC-02: (0) reuses UC-01 rows 0.1–0.2, (2) rows 2.1–2.3, (4) row 4, (5) rows 5.1–5.3.	5.2.1, 5.2.2

Check	Result (pass, or what failed)	Fixed in
K5 Every analyzed use case traces to at least one functional requirement	Pass. UC-01 traces to FR-03, FR-04, FR-05, FR-06, FR-08, FR-12, and UC-02 to FR-04, FR-06, FR-07, FR-08, matching their trace rows in 4.2. Finding: 4.1 lists duplicate detection as non-trivial for UC-01, but no requirement asks for it and 4.2 has no step for it, so 5.2.1 does not model it.	Open question Q4 in section 11
K6 The last messages of each diagram make the use case postcondition true	Pass. UC-01: 5.1 creates the record, 5.4 links it to the student and 5.5 shows it. In A2, A2.4 removes it and A2.5 shows the list without it; a cancel at A2.3 leaves records unchanged. UC-02: 5.2 and 5.3 change the history and status together, and 6 shows both; A1 changes nothing.	—